

004 — The Brain as an Information Job Shop

The brain may be understood not simply as a biological computer, but as an evolving Information Job Shop that continuously manufactures information from perception, memory, reasoning, and interaction.

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with chatGPT (OpenAI)

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Repository: <https://github.com/sizhet/Information-Job-Shop-IJS>

Introduction

Modern neuroscience has revealed extraordinary complexity inside the brain. However, despite remarkable progress, there remains no widely accepted engineering model describing how perception, memory, reasoning, imagination, planning, and decision making cooperate as one continuously operating computational system.

This repository proposes a simple engineering hypothesis.

Instead of viewing the brain primarily as a neural network or a collection of neurons,

the brain may also be viewed as an **Information Job Shop (IJS)**.

Its primary function is not merely to process signals.

Its primary function is to continuously manufacture useful information for survival, adaptation, learning, and future action.

The Brain Does Not Process Pixels

Human vision illustrates an important observation.

Each eye receives a different image.

Neither image perfectly represents the external world.

Nevertheless,

people normally perceive a single coherent scene.

This suggests that the brain is unlikely to perform exhaustive pixel-by-pixel reconstruction.

Instead,

it appears to construct higher-level information objects.

The computational workpiece is therefore not raw sensory data.

It is an evolving Information Job.

Information Jobs Everywhere

During daily life, the brain continuously manufactures many different Information Jobs.

Examples include:

- understanding a scene,
- recognizing an object,
- planning a route,
- solving a mathematical problem,
- recalling a memory,
- preparing a conversation,
- imagining future events,
- generating scientific hypotheses.

Each Information Job follows its own runtime process.

Some complete within milliseconds.

Others continue evolving for years.

Runtime Manufacturing

Unlike traditional manufacturing, brain processing is highly adaptive.

During execution,

an Information Job may continuously:

- retrieve memories,
- activate associations,
- generate new questions,
- revise previous conclusions,
- request additional evidence,
- modify its own objectives,
- recruit additional cognitive resources,
- postpone execution until more information becomes available.

Information manufacturing therefore becomes an ongoing runtime activity rather than a predefined computational sequence.

Conditional Triggering Networks

One of the most important characteristics of biological intelligence is the existence of enormous numbers of conditional triggering relationships.

Examples include:

- danger triggers escape,
- hunger triggers food search,

- curiosity triggers exploration,
- uncertainty triggers reasoning,
- inconsistency triggers verification,
- opportunity triggers planning.

These triggers continuously redirect Information Jobs toward different computational resources.

Instead of static workflows,
the brain operates through dynamic runtime routing.

The resulting processing routes continuously evolve during execution.

Dreaming as Information Manufacturing

Dreams provide another interesting observation.

Dream imagery can be generated rapidly while combining memories, objects, people, locations, and imagined events that have never physically occurred together.

This behavior is difficult to explain as simple pixel reconstruction.

It becomes more understandable if dreams manufacture Information Jobs composed of reusable semantic structures.

Rendering detailed imagery may therefore represent a later stage of runtime processing rather than the primary computational mechanism.

The Brain as an Information Manufacturing Ecosystem

Unlike traditional factories,

the brain does not manufacture isolated products.

Multiple Information Jobs coexist simultaneously.

They compete for attention.

They cooperate.

They exchange intermediate information.

They influence each other's priorities.

They continuously interact with the external environment.

Consequently,

the brain resembles an evolving Information Manufacturing Ecosystem more than a fixed production line.

Information Jobs become active computational entities participating in a continuously evolving cognitive society.

Relation to Artificial Intelligence

Modern AI systems are gradually exhibiting similar characteristics.

LLMs generate new prompts.

Agents create subtasks.

Tools contribute specialized capabilities.

Humans participate in collaborative reasoning.

Future Brain Units may become specialized Information Manufacturing Cells. Although today's AI systems remain far simpler than biological intelligence, Information Job Shop provides a common engineering language capable of describing both.

Rather than copying biology directly, future AI may progressively develop computational architectures exhibiting similar organizational principles.

Engineering Perspective

The purpose of this discussion is not to claim that the brain literally implements an Information Job Shop.

Instead,

Information Job Shop is proposed as an engineering abstraction.

Engineering abstractions are valuable because they organize complexity into understandable computational principles.

Whether implemented by neurons, silicon processors, distributed agents, or future Brain Units, the central question remains the same:

How are Information Jobs manufactured?

Key Takeaways

- The brain appears to process structured information rather than raw sensory data alone.
- Many cognitive activities can be interpreted as Information Jobs with independent runtime lifecycles.
- Conditional Triggering Networks dynamically route Information Jobs during execution.
- Dreams illustrate rapid semantic information manufacturing rather than simple image reconstruction.
- The brain resembles an evolving Information Manufacturing Ecosystem composed of interacting Information Jobs.
- Information Job Shop offers an engineering abstraction that naturally connects biological intelligence with future AI architectures.

"Perhaps intelligence is not defined by neurons or silicon, but by the continuous manufacturing of information through evolving Information Jobs."

